

Ecommerce Behavior Analysis

1. Project Overview

This project analyzes eCommerce sales performance using transactional data collected from multiple online platforms including Amazon, Jumia, and Souq. The goal is to uncover insights into revenue trends, product sales, customer ratings, and category performance to support strategic business decisions.

2. Dataset Summary

- Rows: 10,000
- Columns: 11
- Key Features:
 - Customer demographics (Order id, City, Rating, Reviews,)
 - Purchase details (Product, Category, Price, Quantity, Order date)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for summary statistics.

	OrderID	Product	Category	Brand	Platform	City	Price	Quantity	TotalAmount	Rating	Reviews	OrderDate
count	10000	10000	10000	10000	10000	10000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000
unique	10000	10	5	10	3	6	NaN	NaN	NaN	NaN	NaN	361
top	ORD100000	Samsung Galaxy A14	Electronics	Samsung	Souq	Cairo	NaN	NaN	NaN	NaN	NaN	02-12-2024
freq	1	1043	3028	1043	3371	1705	NaN	NaN	NaN	NaN	NaN	46
mean	NaN	NaN	NaN	NaN	NaN	NaN	10027.986331	3.006200	30199.572439	3.005592	2520.072200	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	5728.261196	1.415472	23861.789828	1.154845	1443.849734	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	105.590000	1.000000	113.550000	1.000000	0.000000	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	4999.862500	2.000000	11066.910000	2.010000	1275.000000	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	10063.600000	3.000000	23671.630000	3.005000	2537.500000	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	14938.430000	4.000000	44648.580000	4.000000	3767.000000	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	19999.340000	5.000000	99993.800000	5.000000	5000.000000	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the columns.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Data Consistency Check:** Verified brand column values, if **Generic** and **generic** were redundant.
- **Identifying Invalid Data Points:** Verified rating column values, if rating having greater than **5**.
- **Feature Engineering:**
 - Created **month** column by binning customer order date.
 - Created **year** column from customer order date.
 - Created **day_name** column from customer order date
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by platform** – Compared total revenue generated by each platform Souq, Jumia, Amazon.

	platform	revenue
▶	Souq	101920977.13
	Jumia	101697072.5
	Amazon	98377674.76

2. **Revenue by top selling categories** – Compared total revenue by category with number of quantities sold.

	category	quantity_sold	revenue
▶	Electronics	9036	92327183.77
	Fashion	8980	91504159.73
	Computers	5892	57542997.56
	Wearables	3111	31420413.2
	Accessories	3043	29200970.13

3. **Revenue by cities** – Compared total revenue by cities with number of quantities sold, ordered by revenue.

	city	quantity_sold	revenue
▶	Cairo	5159	52522076.73
	Casablanca	5164	52382993.51
	Riyadh	5022	50364375.88
	Dubai	4946	49177483.79
	Alexandria	4967	48844458.99
	Giza	4804	48704335.49

4. **Product & Rating performance** – Compared product by total rating and review count.

	product	reviews	total_rating
▶	Bluetooth Headset	2632079	3083.36
	Xiaomi Redmi Note 12	2609353	3076.25
	Samsung Galaxy A14	2604628	3176.58
	T-shirt Cotton	2534748	2939.91
	HP Pavilion Laptop	2526089	3038.51
	Smartwatch FitPro	2518191	3090.3
	Nike Air Max	2505699	2972.13
	Adidas Running Shoes	2464191	2873.74
	Dell Inspiron 3501	2455185	2878.45
	iPhone 13	2350559	2926.69

5. **Higher rating correlate with sales** – .Compared sales and revenue across rating group.

	rating_group	total_quantity_sold	total_revenue
▶	4 - 5	7525	75124791.13
	3 - 4	7555	76394356.65
	0 - 3	14982	150476576.61

6. **Average order value by category** – Compared average order value across category.

	category	avg_order_value
▶	Fashion	30809.48
	Wearables	30804.33
	Electronics	30491.14
	Computers	29463.9
	Accessories	28378.01

7. **Highest sales Month** – Identified the month with the highest total sales.

	month	total_amount
▶	12	21946736.69

8. **Top Revenue Products by Month** – Found the highest revenue-generating products for each category by month.

	category	product	month	total_amt	d_rank
▶	Accessories	Bluetooth Headset	5	2911984.3299999996	1
	Computers	Dell Inspiron 3501	1	2749669.32	1
	Computers	HP Pavilion Laptop	7	3433654.83	1
	Electronics	iPhone 13	9	2892540.010000001	1
	Electronics	Samsung Galaxy A14	9	2973868.5600000005	1
	Electronics	Xiaomi Redmi Note 12	2	3274380.23	1
	Fashion	Adidas Running Shoes	3	3041172.7199999997	1
	Fashion	Nike Air Max	5	3118184.7400000016	1
	Fashion	T-shirt Cotton	9	3018444.3600000013	1
	Wearables	Smartwatch FitPro	4	2903501.7700000005	1

9. **Orders by weekday** – Compared the number of orders across weekdays.

	day_name	no_of_orders
▶	Monday	1394
	Tuesday	1477
	Wednesday	1489
	Thursday	1428
	Friday	1371
	Saturday	1385
	Sunday	1456

10. **Highest Order Day** – Identified the weekday with the highest number of orders.

	day_name	no_of_orders
►	Wednesday	1489

11. **Revenue by Platform**– Compared revenue and revenue contribution by platform.

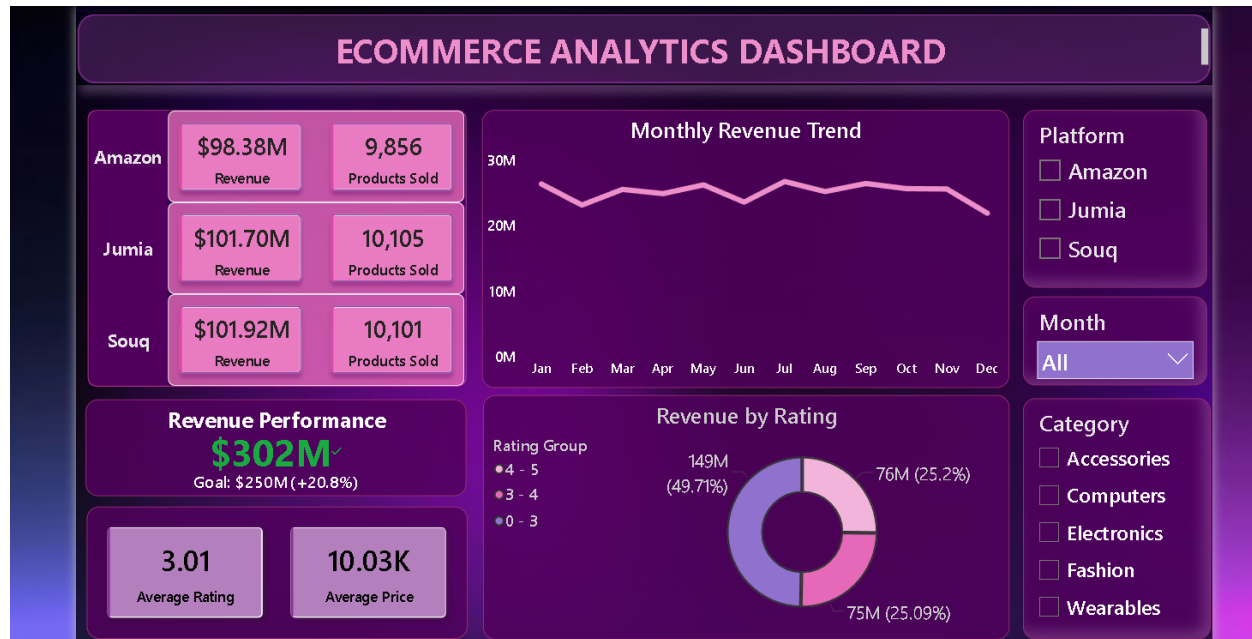
	platform	revenue	percentage
►	Souq	101920977.13	33.75
	Jumia	101697072.5	33.68
	Amazon	98377674.76	32.58

12. **Top 5 Brands by Revenue** – Identified the top 5 brands based on total revenue.

	brand	revenue
►	Xiaomi	33772598.37
	FitPro	31420413.2
	Nike	30727637.51
	Generic	30507500.4
	Adidas	30269021.82

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Target Low performing Markets** – Focus marketing campaigns on low-performing cities to improve customer reach and boost sales growth.
- **Promote Higher rated Products** – Promote highly rated products more aggressively, as customer rating are linked with higher sales performance.
- **Review Discount Policy** – Balance sales boosts with margin control, ensure profitable grow.
- **Product Quality Strategy** – Focus on improving low-performing categories and products through better pricing, promotions, discounts and offers and customer feedback analysis.
- **Product Positioning** – Highlight top-rated and best-selling products in brands to boost profitability.
- **Platform Performance** – Focus efforts on using platform-wise revenue insights to invest more in the best markets like Souq and Jumia.